Module 3: Probability and Distributions Review

Connect the experiment, the model, and the interpretation

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Purpose and submission

Prepare a short Quarto report with the five sections below. Give the event in notation, justify the method and assumptions, show reproducible R code, and interpret every result with its units or reference group. Keep instructor answers separate.

These are the cases introduced in Lessons 1–5. Return to their English–Arabic introductions for the business settings; this review introduces no new case or observations.

1. Define the probability experiment

Use the Restaurant CSV, with uniform selection of one of its 300 records. Define A as a Very Good rating and B as a price from USD 20 to less than USD 30.

1. Find $P(A \cup B)$ and $P(A \cap B)$.
2. Explain why selecting uniformly among the three rating names would be a different experiment.
3. State why these probabilities do not establish the proportion in today's restaurant market.

2. Choose the denominator

1. Calculate $P(A \mid B)$ and $P(B \mid A)$; explain the difference.
2. Determine whether these events are independent in the finite selection experiment.
3. Using the two-supplier inputs from Lesson 2, find $P(S_1 \mid D)$ and distinguish it from $P(D \mid S_1)$.

3. Interpret a discrete distribution

Use the six published DiCarlo frequency cells in Lesson 3.

1. Construct the PMF and show that its masses sum to 1.
2. Find $P(X \geq 3)$, the expected daily count, and the model SD.
3. Explain why 1.50 is neither the mode nor a possible one-day count.
4. Distinguish the published aggregate frequencies from an individual-day dataset.

4. Select a count model

1. For Martin Clothing Store, justify the binomial model and find the probability of at least two purchases among the next three customers.
2. For the arrivals example, state the Poisson-process assumptions and calculate the probability of no arrivals in three minutes.
3. Explain why using mean count 10 in that three-minute calculation is incorrect.
4. Explain why simulation under either model does not verify its assumptions against real operations.

5. Separate area, height, and cutoff

1. Under the flight model, find $P(128 \leq T \leq 136)$ and explain why the density height is a different quantity.
2. Under the Grear normal model, find $P(X > 40{,}000)$ and the 10th-percentile lifetime in miles.
3. Explain why a z-score alone does not justify a normal model.
4. State which raw observations are unavailable and what that prevents you from checking.

Review checklist

Criterion	Evidence in the report
Clear experiment	Outcome, event, and reference group
Suitable model	Variable type, parameters, units, and assumptions
Correct event	Exact / at most / at least / more than translated correctly
Reproducible calculation	CSV for observations; cited constants for published model inputs
Sound interpretation	Probability versus density, expectation versus realization
Honest limits	Historical data, model assumptions, unavailable raw observations

This completes the selected Module 3 sequence. Proceed to Module 4: Sampling and estimation when its lessons are available, or return to the Module 3 overview.